

Genetic Algorithm – Mathematical Formulation

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1 What GA Does

GA maintains a population of 80 schedules (chromosomes) and evolves them over up to 300 generations using selection, crossover, and mutation. *Code: src/algorithms/genetic_algorithm.py, function solve_genetic_algorithm()*

2 Chromosome Encoding

A chromosome is a list of T integers (one per trip):

$$C = [u_1, u_2, \dots, u_T], \quad u_i \in \{0, 1, \dots, N - 1\}$$

u_i = index of the unit assigned to trip i (trips sorted by departure_min). *Built in: solve_genetic_algorithm() population initialization*

3 Fitness Function

$$fitness(C) = Cost(C) + 10 \cdot w_5 \cdot V_{time}(C)$$

Lower is better. Timing violations penalized 10x extra. *Computed by: compute_cost() then penalty applied in evaluate loop*

4 Selection

Tournament selection with $k = 5$: pick 5 random chromosomes, keep the best. *Code: tournament_select() inside solve_genetic_algorithm()*

5 Crossover (PMX, probability 0.85)

Take two cut points $a < b$. Swap genes between positions a and b :

$$Child_1[a : b] = Parent_2[a : b], \quad Child_1[rest] = Parent_1[rest]$$

After crossover: call repair_chromosome() to fix infeasible assignments. *Code: crossover block + repair_chromosome() from src/Utils/feasibility.py*

6 Mutation (probability 0.10 per gene)

Randomly replace a gene with a different valid unit index. *Code: mutation block in solve_genetic_algorithm()*

7 Elitism

Top 2 chromosomes always survive to next generation. *Code: elitism copy at start of each generation loop*

8 Why GA Underperforms Here

Crossover breaks physical continuity – unit assigned in Parent A is now at a different location due to Parent B assignments. The result is trains that teleport between stations. Repair fixes some but not all. Cost at 60 trips: 7,910.6 (worse than greedy at 5,696.9).